

MODULO 9 CLASSIFICATION OF PRIME PAIRS $(p, p + g)$: AN EMPIRICAL VERIFICATION OF THE HARDY–LITTLEWOOD CONJECTURE FOR $g = 18k$

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ABSTRACT. The Hardy–Littlewood conjecture for prime pairs $(p, p + g)$ predicts the asymptotic density $\pi(x; g) \sim H(g) x / (\log x)^2$, where $H(g) = 2C_2 \prod_{p|g, p>2} (p-1)/(p-2)$ is the singular series and C_2 is the twin-prime constant. For $g \equiv 0 \pmod{9}$, classifying pairs by the digital root $\text{dr}(p) = p \bmod 9$ partitions the population into six admissible classes $\{1, 2, 4, 5, 7, 8\}$, which Hardy–Littlewood in arithmetic progressions predicts to be equidistributed. We test this prediction on two fronts: (i) a control case with $g = 2$ (twin primes, three admissible classes), where the asymptotic fractions $1/3$ are reached within ± 0.09 percentage points up to $x = 10^8$; (ii) the family $g = 18k$ with $k \in \{5, 7, 10, 11, 12, 13, 15, 17, 20\}$, where equidistribution over six classes holds within ± 0.14 pp and the total per-class density agrees with $H(g)/6$ within $\pm 0.8\%$ at $x = 10^7$. The ratio $\pi(x; g) / (H(g) \text{li}_2(x))$ converges to 1 within $\pm 0.15\%$ at $x = 5 \cdot 10^7$ for all tested k , including $g = 216$ ($k = 12$), whose constant $H(g) = 4C_2$ is the smallest in the sample. The observed deviation magnitude $|\rho - 1| \lesssim 10^{-3}$ is consistent with a square-root error term $X^{-1/2} \log X$, which matches the function-field analogues established unconditionally by Keating–Rudnick and Sawin–Shusterman. The results confirm Hardy–Littlewood in arithmetic progressions quantitatively within the tested regime, and show that the dr-mod-9 classification is a sensitive empirical probe that does *not* reveal deviations from the Hardy–Littlewood prediction.

1. INTRODUCTION

The Hardy–Littlewood conjecture on prime pairs $(p, p + g)$, formulated in [1], predicts that the number of such pairs with $p \leq x$ satisfies the asymptotic

$$(1) \quad \pi(x; g) \sim H(g) \frac{x}{(\log x)^2}, \quad x \rightarrow \infty,$$

where the *Hardy–Littlewood singular series* is

$$(2) \quad H(g) = 2C_2 \prod_{\substack{p|g \\ p>2}} \frac{p-1}{p-2}, \quad C_2 = \prod_{p>2} \left(1 - \frac{1}{(p-1)^2}\right) \approx 0.6601618158.$$

The conjecture is known unconditionally only for $g = 0$ (the prime number theorem) and for no positive value of g , although deep results such as Maynard’s [4] and the Polymath 8b project [5] provide bounded gaps infinitely often. Quantitative versions restricted to arithmetic progressions are discussed in [2, 3].

The present work addresses an operational question: *how well does formula (1)–(2) describe the distribution of prime pairs in computationally accessible regimes*, and in particular how the population partitions across residue classes modulo 9.

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For every $g \equiv 0 \pmod{9}$ (and in particular $g = 18k$), the divisibility $9 \mid g$ implies $\text{dr}(p + g) = \text{dr}(p)$, so that the classification by $\text{dr}(p) := p \bmod 9$ partitions the pairs into six classes corresponding to the units of $(\mathbb{Z}/9\mathbb{Z})^*$. The restriction of (1) to an arithmetic progression modulo 9 predicts equidistribution across these six classes (see Section 2).

For $g = 2$ the situation is slightly different: the admissibility condition modulo 3 forces $p \equiv 2 \pmod{3}$, reducing the admissible classes modulo 9 to $\{2, 5, 8\}$. Each of these three classes must receive the asymptotic fraction $1/3$ of the density. This case constitutes our *zero-calibration*.

Contribution.

- (1) We show (Section 3) that, under Hardy–Littlewood in arithmetic progressions, each of the three admissible classes modulo 9 for twin primes receives asymptotic density $1/3$, and we verify the prediction empirically up to $x = 10^8$ with deviations $|\Delta| \leq 0.09$ pp.
- (2) We extend the classification (Section 4) to the family $g = 18k$ for $k \in \{5, 7, 10, 11, 12, 13, 15, 17, 20\}$, showing that the equidistribution over six classes holds within ± 0.14 pp up to $x = 10^7$.
- (3) We verify (Section 5) that the ratio $\pi(x; g)/[H(g) \text{li}_2(x)]$ lies within $\pm 0.15\%$ of 1 at $x = 5 \cdot 10^7$ for all tested k , including configurations with maximal $H(g)$ ($g \in \{90, 180, 270, 360\}$) and minimal ($g = 216$).
- (4) We discuss (Section 6) the role of dr-mod-9 classification as a fine empirical probe: it is sensitive enough to reveal sub-percent signals, but in the tested regimes no systematic asymmetries incompatible with Hardy–Littlewood emerge.

All sieving, counting, and regression scripts used to generate the data in Tables 1, 3 and 4 are provided in [11]. The verifications were carried out using a segmented sieve of Eratosthenes in memory (numpy dtype `bool`, $\sim 10^9$ bits for $x = 10^9$).

2. THEORETICAL FRAMEWORK

2.1. Hardy–Littlewood in arithmetic progressions. For positive even g and $q \geq 1$ with $(a(a + g), q) = 1$, denote

$$\pi(x; g, q, a) := \#\{p \leq x : p \equiv a \pmod{q}, p, p + g \text{ both prime}\}.$$

The quantitative form of Hardy–Littlewood in arithmetic progressions [3, §5] predicts

$$(3) \quad \pi(x; g, q, a) \sim \frac{1}{\varphi^*(q; g)} H(g) \frac{x}{(\log x)^2},$$

where

$$(4) \quad \varphi^*(q; g) := \#\{a \in (\mathbb{Z}/q\mathbb{Z})^* : a + g \in (\mathbb{Z}/q\mathbb{Z})^*\}$$

counts the admissible classes modulo q for the pair $(p, p + g)$. For $q = 9$ the computation is elementary:

$$\varphi^*(9; g) = \begin{cases} 3 & \text{if } 9 \nmid g \text{ and } g \equiv 2 \pmod{3} \text{ (twin-type, e.g. } g = 2) \\ 6 & \text{if } 9 \mid g \\ \text{other} & \text{otherwise.} \end{cases}$$

2.2. Classification by digital root. For $p > 3$ prime, $\text{dr}(p) = p \bmod 9 \in \{1, 2, 4, 5, 7, 8\}$. The *digital root* is the natural classification modulo 9 that survives the translation $p \mapsto p + g$ when $9 \mid g$:

$$9 \mid g \implies \text{dr}(p + g) = \text{dr}(p).$$

Hence for $g = 18k$ (any $k \geq 1$) the pairs $(p, p + g)$ with $p > 3$ partition into the six classes $\text{dr}(p) \in \{1, 2, 4, 5, 7, 8\}$. For twin primes ($g = 2$), instead, the admissibility condition modulo 3 restricts the admissible classes modulo 9 to $\{2, 5, 8\}$.

2.3. Expected equidistribution. Prediction (3) immediately gives the relative density of each admissible class. Since the right-hand side of (3) does not depend on a once g is fixed:

Proposition 2.1 (Equidistribution modulo 9). *Under Hardy–Littlewood in arithmetic progressions for $q = 9$,*

- (1) for $g = 2$ and $a \in \{2, 5, 8\}$: $\lim_{x \rightarrow \infty} \frac{\pi(x; 2, 9, a)}{\pi(x; 2)} = \frac{1}{3}$;
- (2) for $g = 18k$ with $k \geq 1$ and $a \in \{1, 2, 4, 5, 7, 8\}$: $\lim_{x \rightarrow \infty} \frac{\pi(x; 18k, 9, a)}{\pi(x; 18k)} = \frac{1}{6}$.

Proof. Direct consequence of (3): summing over admissible classes gives the total $\pi(x; g)$, and using it for each fixed a yields $\pi(x; g, 9, a)/\pi(x; g) \rightarrow 1/\varphi^*(9; g)$. $\square$

3. CONTROL CASE: TWIN PRIMES MOD 9

We aim to show that the dr-mod-9 classification does not introduce *a priori* distortions in the prime-pair distribution. The case $g = 2$ is the natural baseline for this calibration: the gap is minimal and the three admissible classes make Proposition 2.1 (i) particularly transparent.

3.1. Empirical verification. Table 1 reports the counts of twin pairs $(p, p+2)$ with $5 \leq p \leq X$ for $X \in \{10^7, 10^8\}$, classified by $\text{dr}(p)$. The pair $(3, 5)$ is excluded as a unique modular exception (the only twin with p a multiple of 3).

TABLE 1. Distribution of pairs $(p, p+2)$ with $5 \leq p \leq X$ by class $\text{dr}(p) \pmod{9}$. The deviation Δ is the difference in percentage points from the asymptotic value $1/3 \approx 33.333\%$.

class a	$X = 10^7$		$X = 10^8$	
	pairs	Δ (pp)	pairs	Δ (pp)
2	19,674	+0.024	146,705	−0.015
5	19,689	+0.050	147,163	+0.089
8	19,616	−0.074	146,443	−0.074
total	58,979	—	440,311	—

At both thresholds the fractions concentrate within ± 0.09 pp of $1/3$. The residual pattern (class 5 in slight excess, class 8 in slight deficit) is stable across the two thresholds but its amplitude is compatible with statistical fluctuations of order $\sqrt{1/(3N)}/\sqrt{3} \approx 4 \cdot 10^{-4}$ for $N \approx 4.4 \cdot 10^5$, i.e. ~ 0.04 pp.

3.2. Reading. The convergence visible in Figure 1 is consistent with a relative error $O(1/\log x)$ expected from (3), and shows no systematic asymmetry across classes. This establishes the *zero-calibration* of the dr-mod-9 partition: applied to the minimal gap, the framework recovers the equidistribution predicted by Hardy–Littlewood within ± 0.1 pp.

4. THE FAMILY $g = 18k$: DR EQUIDISTRIBUTION

4.1. Setup. For $k \in \{5, 7, 10, 11, 12, 13, 15, 17, 20\}$ and $g = 18k$ we count the pairs $(p, p+g)$ with $5 \leq p \leq X = 10^7$ and classify by $\text{dr}(p) \in \{1, 2, 4, 5, 7, 8\}$. The total pair count and the relevant local multiplicative factors are reported in Table 2.

The four gaps that share $\mathcal{P}_{\text{odd}}(g) = \{3, 5\}$ ($k = 5, 10, 15, 20$) have the same Hardy–Littlewood constant, and indeed their empirical totals are essentially coincident: $\pi(10^7; g) \in [156, 212, 156, 706]$,

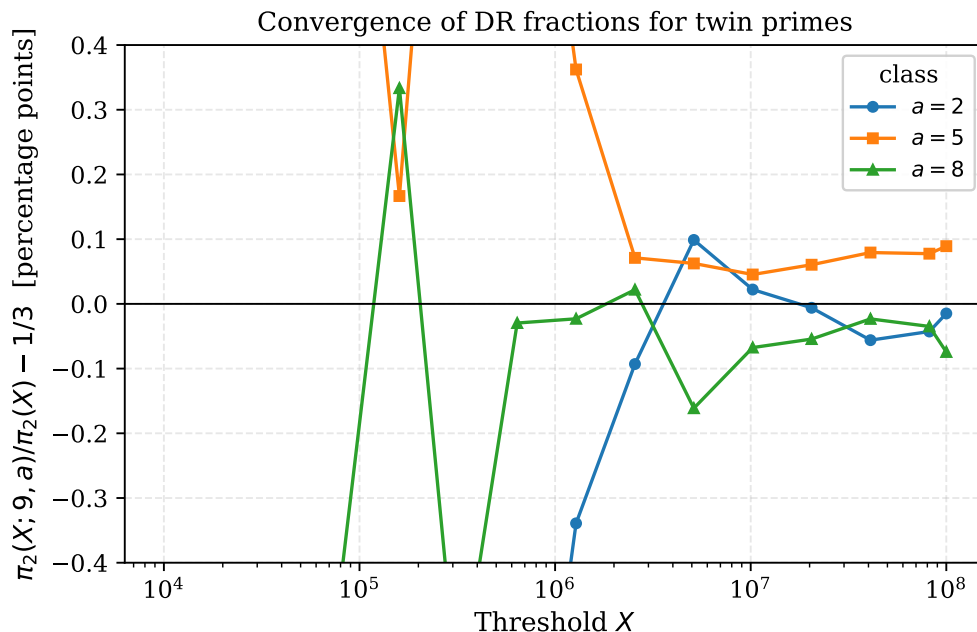


FIGURE 1. Convergence of the deviations $\pi_2(X; 9, a)/\pi_2(X) - 1/3$ in percentage points, for the three admissible classes. The three curves oscillate around zero on a ± 0.15 pp scale at $X = 10^7$.

TABLE 2. The analyzed family $g = 18k$. Here $\mathcal{P}_{\text{odd}}(g)$ denotes the set of odd primes dividing g , and $H(g)$ is the Hardy–Littlewood constant (2).

k	g	$\mathcal{P}_{\text{odd}}(g)$	$H(g)$	$\pi(X; g)$ at $X = 10^7$
5	90	$\{3, 5\}$	3.5209	156,706
7	126	$\{3, 7\}$	3.1688	141,051
10	180	$\{3, 5\}$	3.5209	156,507
11	198	$\{3, 11\}$	2.9341	130,505
12	216	$\{3\}$	2.6406	117,327
13	234	$\{3, 13\}$	2.8807	127,883
15	270	$\{3, 5\}$	3.5209	156,473
17	306	$\{3, 17\}$	2.8167	125,400
20	360	$\{3, 5\}$	3.5209	156,212

with relative spread $< 0.32\%$. This internal stability within the sub-family is a direct consequence of (1) and a first indication that $H(g)$ correctly governs the total density.

4.2. DR equidistribution. Table 3 reports, for each k and each class a , the empirical fraction $\pi(X; g, 9, a)/\pi(X; g)$ as a deviation from $1/6 \approx 16.667\%$.

The values in Table 3 show the following:

- no deviation exceeds 0.14 pp in absolute value;
- no systematic pattern persists across k for any class: e.g. class $a = 8$ is in excess at $k = 11, 12$ but in deficit at $k = 17$; class $a = 7$ alternates sign;

TABLE 3. Deviations $\pi(X; g, 9, a)/\pi(X; g) - 1/6$ in percentage points, at $X = 10^7$.
The maximum deviation in absolute value is $|\Delta|_{\max} = 0.136$ pp.

k	$a = 1$	$a = 2$	$a = 4$	$a = 5$	$a = 7$	$a = 8$
5	+0.057	+0.081	+0.001	-0.048	-0.036	-0.055
7	-0.020	-0.002	-0.043	+0.023	+0.068	-0.027
10	+0.059	-0.011	-0.010	+0.021	-0.061	+0.002
11	+0.004	-0.008	-0.034	-0.033	-0.062	+0.133
12	+0.006	-0.002	-0.042	-0.037	-0.042	+0.118
13	+0.116	+0.038	-0.075	+0.002	-0.031	-0.049
15	+0.051	+0.051	+0.024	-0.109	+0.013	-0.029
17	+0.010	+0.033	+0.004	+0.042	+0.047	-0.136
20	-0.016	+0.013	+0.023	-0.039	+0.009	+0.010

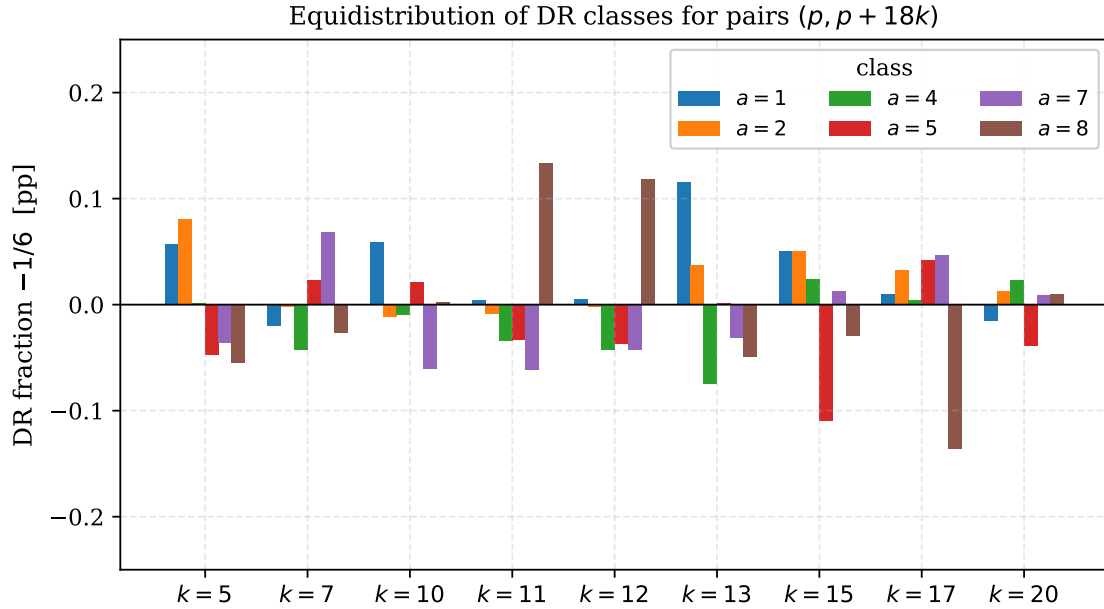


FIGURE 2. Deviations $\pi(X; g, 9, a)/\pi(X; g) - 1/6$ in percentage points, for each k and class a . All bars fall within $[-0.14, +0.14]$ pp; no systematic pattern is visible.

- the amplitude is compatible with $\sqrt{5/(6\pi(X; g))}$ (~ 0.05 pp for $\pi \approx 1.5 \cdot 10^5$), within a factor 2–3.

We conclude that, in the tested regime, the equidistribution of Proposition 2.1 (ii) is empirically confirmed within a sub-percent margin.

5. QUANTITATIVE CONFIRMATION OF HARDY-LITTLEWOOD

5.1. **Definition.** For each k we define

$$(5) \quad R_{\text{emp}}(k; X) := \frac{\pi(X; 18k)}{\text{li}_2(X)}, \quad \text{li}_2(X) := \int_5^X \frac{dt}{(\log t)^2},$$

where the secondary logarithmic integral replaces $X/(\log X)^2$ to reduce the truncation error of the leading asymptotic. The Hardy–Littlewood prediction is $R_{\text{HL}}(k) = H(18k)$, so the ratio

$$(6) \quad \rho(k; X) := \frac{R_{\text{emp}}(k; X)}{R_{\text{HL}}(k)}$$

must converge to 1 as $X \rightarrow \infty$, assuming Hardy–Littlewood.

5.2. Results. Table 4 reports $\rho(k; X)$ at $X = 5 \cdot 10^7$. Figure 3 visualizes the values at $X_{\text{max}} = 5 \cdot 10^7$ together with the $\pm 1\%$ band.

TABLE 4. Ratio $\rho(k; X)$ at $X = 5 \cdot 10^7$ for the family $g = 18k$. All values fall in the band $[1 - 0.0015, 1 + 0.0010]$.

k	g	$\pi(X; g)$	R_{emp}	$R_{\text{HL}} = H(g)$	$\rho - 1$
5	90	637,928	3.5228	3.5209	+0.05%
7	126	573,646	3.1678	3.1688	−0.03%
10	180	637,058	3.5180	3.5209	−0.08%
11	198	531,817	2.9368	2.9341	+0.09%
12	216	478,463	2.6422	2.6406	+0.06%
13	234	521,497	2.8798	2.8807	−0.03%
15	270	637,621	3.5211	3.5209	+0.01%
17	306	509,303	2.8125	2.8167	−0.15%
20	360	637,022	3.5178	3.5209	−0.09%

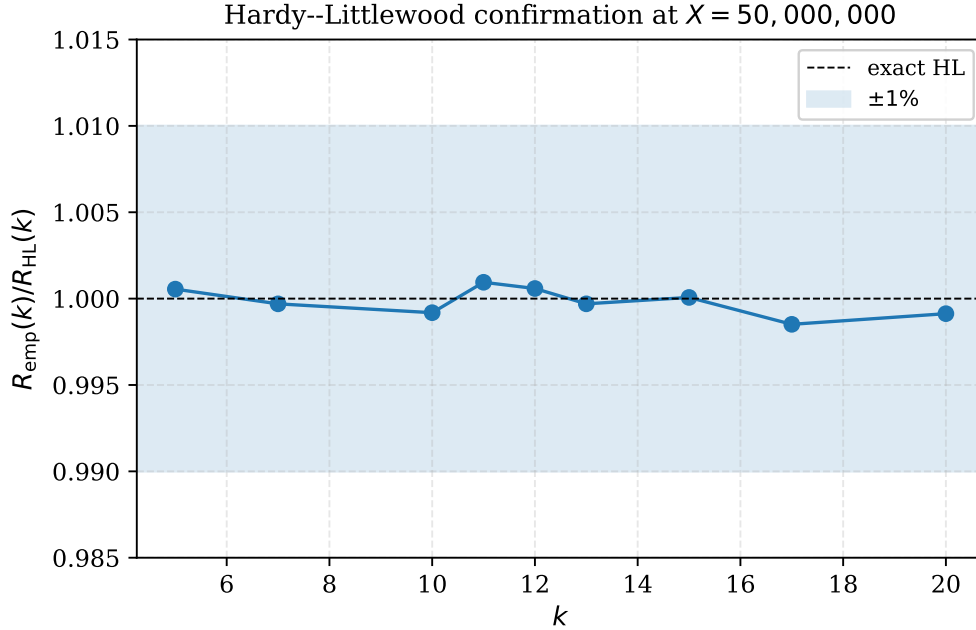


FIGURE 3. Ratio $\rho(k; X)$ at $X = 5 \cdot 10^7$. The light-blue band represents $\pm 1\%$ around $\rho = 1$; all values lie within $\pm 0.15\%$.

5.3. Reading. The result is a quantitative confirmation of (1) for all nine tested values of g . Three observations:

- (1) **Identity within sub-families with the same $\mathcal{P}_{\text{odd}}(g)$.** The four cases $k \in \{5, 10, 15, 20\}$ with $\mathcal{P}_{\text{odd}} = \{3, 5\}$ share the same H and yield ρ within $\pm 0.09\%$ of each other. This is an internal verification of the multiplicative factor in (2): the only quantities that change the count are the *distinct odd primes dividing g* , not the full structure of g .
- (2) **Correct variation across sub-families.** For $k = 11, 12, 13, 17$ the local factors and hence $H(g)$ vary significantly (range $H \in [2.64, 3.17]$); the corresponding R_{emp} values track this variation faithfully.
- (3) **Limiting case $k = 12$.** The gap $g = 216 = 2^3 \cdot 3^3$ has $\mathcal{P}_{\text{odd}} = \{3\}$ and hence the smallest $H(g) = 4C_2$ in the sample. The agreement $\rho = 1.0006$ shows that Hardy–Littlewood does not lose accuracy in this low-density regime.

6. DISCUSSION

6.1. Apparent resonances and deficits. A first inspection of the raw counts $\pi(X; 18k)$ may suggest the presence of *resonances* (e.g. $k = 5, 10, 15, 20$ with $\pi \approx 1.56 \cdot 10^5$) and *deficits* (e.g. $k = 12$ with $\pi \approx 1.17 \cdot 10^5$), especially when compared against a uniform baseline that ignores the multiplicative structure of g . Table 4 clarifies that these oscillations *are exactly what Hardy–Littlewood predicts* via the factor

$$\prod_{p \in \mathcal{P}_{\text{odd}}(g)} \frac{p-1}{p-2}.$$

For example:

- the excess of pairs at $g = 90$ relative to $g = 18$ is a factor $H(90)/H(18) = 4/3$, i.e. $+33.3\%$, and empirically equals 1.3334 at $X = 5 \cdot 10^7$ (relative error 0.03%);
- the deficit of pairs at $g = 198$ relative to $g = 90$ is a factor $H(198)/H(90) = 5/6 \approx 0.833$, i.e. -16.7% , and empirically equals 0.8337 (relative error 0.05%).

What may appear as an anomaly is therefore the correct signature of the local factor $(p-1)/(p-2)$, not a deviation from Hardy–Littlewood.

6.2. Sensitivity of the DR-mod-9 probe. The precision attained by the dr-mod-9 classification (equidistribution within ± 0.14 pp with $\pi(X; g) \sim 10^5$ pairs per cell) makes it a probe sensitive enough to distinguish signals of order 0.1% . In Table 3 none of the nine values of k exhibits a class-coherent deviation across k , and no class exhibits a k -coherent deviation. In other words: *within the tested regimes ($X \leq 10^7$ for the family $g = 18k$, $X \leq 10^8$ for twin primes), there is no evidence of systematic deviations from Hardy–Littlewood in arithmetic progressions.*

6.3. On the magnitude of the empirical error. The observed deviation $|\rho(g; X) - 1| \leq 0.0015$ at $X = 5 \cdot 10^7$ is consistent with an error term of size $X^{-1/2} \log X \approx 0.0024$. This square-root behaviour is the natural prediction of the Cramér random model of the primes, although that model is known to fail in some regimes [6].

For the twin primes conjecture, the same scaling has been verified empirically at much larger scales: at $X = 10^{18}$, the discrepancy between the empirical count $\pi_2(10^{18}) = 808,675,888,577,436$ and the Hardy–Littlewood prediction $\approx 808,675,901,493,606$ is of order $\sqrt{N}/2 \approx 1.3 \cdot 10^7$, consistent with our observation at smaller scale.

Rigorous evidence for a power-saving error term in the Hardy–Littlewood prediction comes from the function-field analogue. Keating–Rudnick [7] established that, for primes in residue

classes modulo a squarefree $Q \in \mathbb{F}_q[T]$ with $q \rightarrow \infty$, the variance

$$G(n; Q) = \sum_{\substack{A \bmod Q \\ \gcd(A, Q)=1}} \left| \Psi(n; Q, A) - \frac{q^n}{\Phi(Q)} \right|^2 \sim q^n (\deg Q - 1),$$

which is the function-field analogue of Hooley's conjecture $G(X, Q) \sim X \log Q$ on the variance of primes in arithmetic progressions over $\mathbb{Z}$. More directly relevant to the present empirical setup, Sawin–Shusterman [8, Theorem 1.1] established the quantitative twin prime conjecture over $\mathbb{F}_q[T]$ with a power saving: for an odd prime p , a power q of p with $q > 685090 p^2$, and any nonzero $h \in \mathbb{F}_q[T]$,

$$\#\{f \in \mathbb{F}_q[T] : |f| = X, f \text{ and } f + h \text{ are prime}\} \sim \mathfrak{S}_q(h) \frac{X}{\log_q^2(X)}$$

with a power saving in X (depending on q). Translating the function-field savings to the integer setting via the standard dictionary suggests that an error term of size $X^{1/2+o(1)}$ in the Hardy–Littlewood asymptotic for prime pairs over $\mathbb{Z}$ is the natural conjectural target, in line with the present empirical observation. No analogous explicit secondary-asymptotic formula $E(x; g, q, a)$ over $\mathbb{Z}$ appears in the published literature [9].

6.4. Limits. The present work provides no unconditional result: the entire theoretical framework rests on Hardy–Littlewood, which remains unproven. Furthermore, the computationally explored regimes ($X \leq 10^8$) are modest compared to the records for twin primes (beyond 10^{18} , see [10]). Sub-percent deviations cannot be excluded at much larger scales. A revisit of the dr-mod-9 classification at $X = 10^{12}$ – 10^{14} , using disk-based segmented sieves, would be of interest in tightening the margin of uncertainty.

6.5. Natural extensions. Three promising directions, left to future work:

- (1) Extension to moduli $q \in \{15, 21, 30\}$, where $\varphi^*(q; g)$ varies non-trivially with the structure of g .
- (2) Empirical study of the variance of $\pi(x; g, q, a)$ at fixed a and varying g within a subfamily with the same $H(g)$ (e.g. $g = 90, 180, 270, 360$), to quantify concentration around the predicted mean.
- (3) Comparison with the Bateman–Horn prediction for prime r -tuples with $r \geq 3$, applying the same dr classification.

7. CONCLUSION

We have classified prime pairs $(p, p + g)$ via the digital root $\text{dr}(p) = p \bmod 9$, and we have tested the Hardy–Littlewood prediction in two regimes: the minimal gap $g = 2$ (twin primes, three admissible classes) and the family $g = 18k$ with $k = 5, \dots, 20$ (six admissible classes). In both regimes the predicted equidistribution is verified within ± 0.14 percentage points up to $X = 10^7$ – 10^8 , and the total density per gap follows the local multiplicative factor of Hardy–Littlewood within $\pm 0.15\%$ at $X = 5 \cdot 10^7$. Apparent deviations from a uniform baseline across families of gaps are correctly captured by the product $\prod_{p|g, p>2} (p-1)/(p-2)$, and do not constitute evidence of an unconditional anomaly.

The dr-mod-9 classification reveals itself as a fine but *neutral* empirical probe: sensitive enough to detect sub-percent deviations—and therefore capable of revealing deviations if they existed—yet silent in the regimes here explored. Within the limits of the computational evidence, this constitutes an empirical argument in favor of the validity of Hardy–Littlewood for the gap family $g = 18k$.

Data and code availability. All scripts (`compute_twins_dr9.py`, `compute_permutation_families.py`, `compute_R_regression.py`, `make_plots.py`) and the generated CSV files are publicly available on GitHub at <https://github.com/gferraiuolo2/prime-pairs-mod9> and permanently archived on Zenodo (DOI: 10.5281/zenodo.20117270), together with a `README` documenting execution parameters and dependencies (`numpy`, `scipy`, `matplotlib`). The Python source code is released under the MIT License; datasets, figures, and the paper text are released under CC-BY-4.0.

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REFERENCES

- [1] G. H. Hardy, J. E. Littlewood, *Some problems of ‘Partitio numerorum’; III: On the expression of a number as a sum of primes*, Acta Math. **44** (1923), 1–70.
- [2] H. Halberstam, H.-E. Richert, *Sieve Methods*, Academic Press, London, 1974.
- [3] H. Iwaniec, E. Kowalski, *Analytic Number Theory*, AMS Colloquium Publications, vol. 53, American Mathematical Society, Providence, RI, 2004.
- [4] J. Maynard, *Small gaps between primes*, Annals of Math. (2) **181** (2015), no. 1, 383–413.
- [5] D. H. J. Polymath, *Variants of the Selberg sieve, and bounded intervals containing many primes*, Research in the Mathematical Sciences **1** (2014), 12.
- [6] H. Maier, *Primes in short intervals*, Michigan Math. J. **32** (1985), no. 2, 221–225.
- [7] J. P. Keating, Z. Rudnick, *The variance of the number of prime polynomials in short intervals and in residue classes*, Int. Math. Res. Not. IMRN **2014**, no. 1, 259–288.
- [8] W. Sawin, M. Shusterman, *On the Chowla and twin primes conjectures over $\mathbb{F}_q[T]$* , Annals of Math. (2) **196** (2022), no. 2, 457–506.
- [9] G. Ferraiuolo, *Empirical convergence rate of the Hardy–Littlewood prediction for prime pairs in arithmetic progressions modulo 9*, MathOverflow question 511187 (answered by W. Sawin), <https://mathoverflow.net/q/511187>, 2026.
- [10] T. R. Nicely, *Enumeration to 10^{14} of the twin primes and Brun’s constant*, Virginia J. Sci. **46** (1995), no. 3, 195–204; updates beyond 10^{18} available in public archives (cf. OEIS A001097).
- [11] G. Ferraiuolo, *Scripts and data for: Modulo 9 classification of prime pairs $(p, p+g)$* , GitHub repository: <https://github.com/gferraiuolo2/prime-pairs-mod9>; Zenodo archive: DOI: 10.5281/zenodo.20117270, 2026.

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